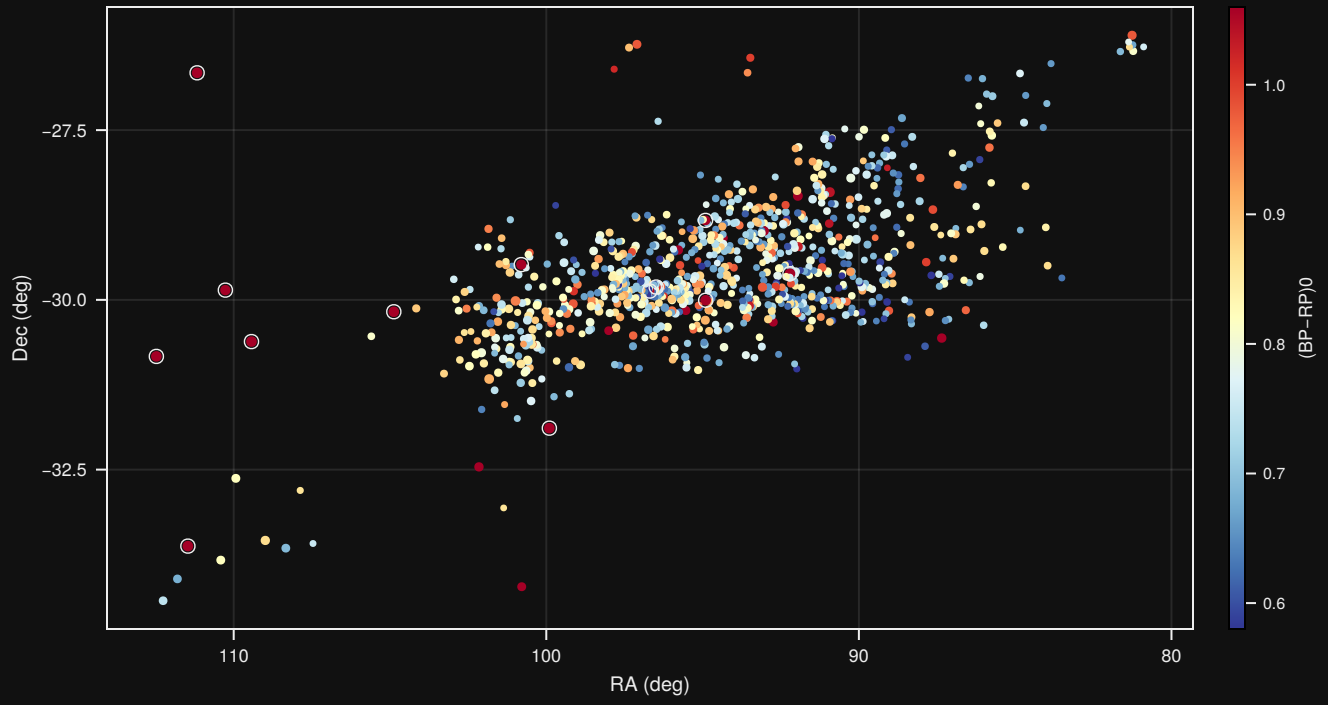
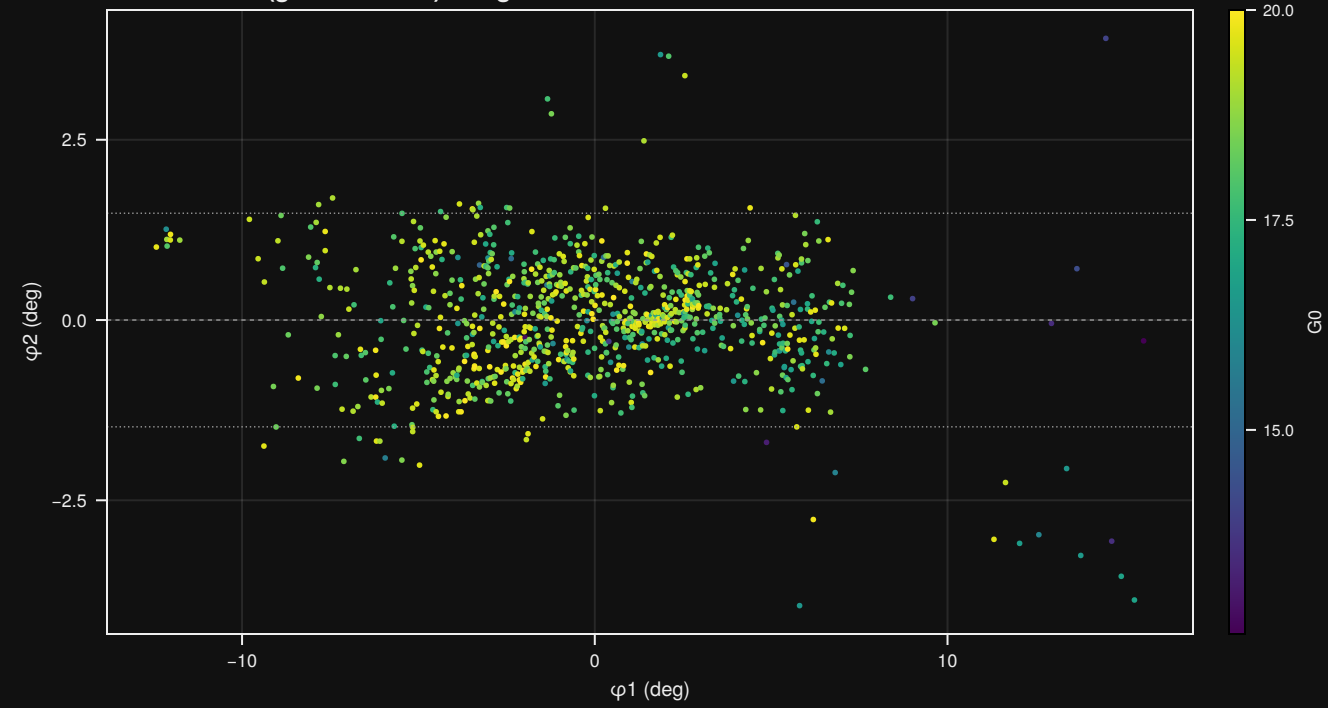
C-12

MEMBERSHIP	
Stars in catalog	929
with Gaia GSP-Phot	491 (53%)
with radial velocity	13
Stream id (sID)	20
GEOMETRY · great-circle fit	
Centroid RA, Dec	94.5, -29.7 deg
Centroid l, b	236.9, -19.7 deg
Pole RA, Dec	109.9, 59.4 deg
Length (φ1, 2–98%)	15.3 deg
Width σ(φ2), robust	0.74 deg
KINEMATICS	
pmRA* med ± σ	-0.50 ± 0.54 mas/yr
pmDE med ± σ	-1.54 ± 0.41 mas/yr
μφ1* med ± σ	-0.27 ± 0.60 mas/yr
μφ2 med ± σ	-1.58 ± 0.41 mas/yr
Parallax (cat.) med	0.160 ± 0.193 mas
Parallax (wtd, ZP-corr.)	0.217 ± 0.011 mas
Distance	d = 4.6 ± 0.2 kpc
Physical length	1.23 kpc
RV median ± σ(MAD)	293.2 ± 28.5 km/s
RV gradient dV/dφ1	5.12 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.6–20.0 (18.7)
(BP−RP)0 median	0.77
GSP-Phot [M/H] (n=491)	-1.16 ± 0.65
GSP-Phot Teff median	5399 K
RUWE > 1.4	2.4 %
PM clusters (largest)	60 (41 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	929 | 929
n_RV catalog | paper	13 | 13
<[Fe/H]> paper	-1.34
Stellar mass (paper)	14000 Msun
Pole offset vs paper	0.0 deg
Remark	Two overlapping structures?

Sky map · catalog members (n = 929) · size ∝ brightness · ring = has RV

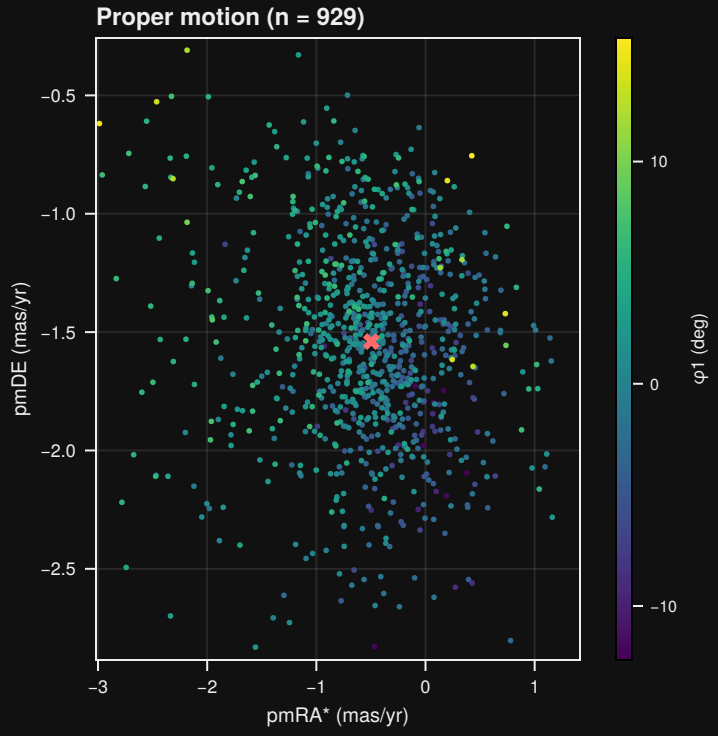


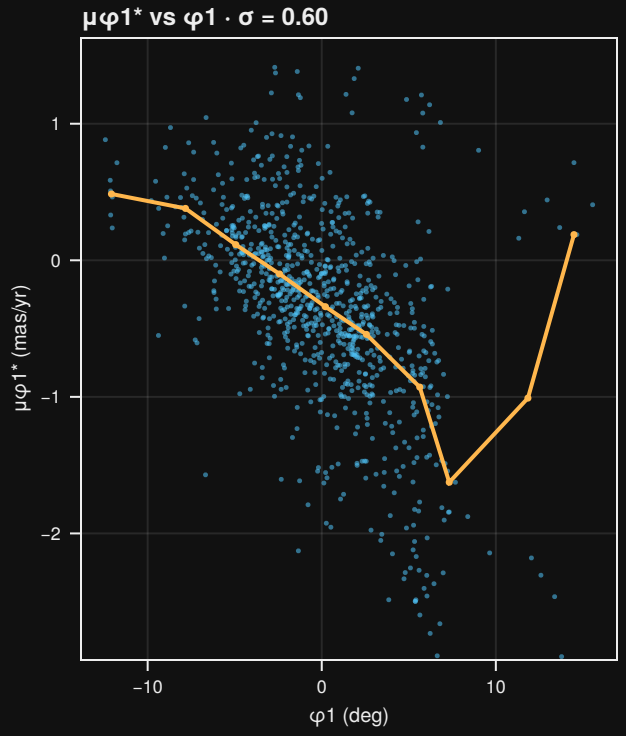
Stream frame (great-circle fit) · length 15.3° · width σ 0.74°

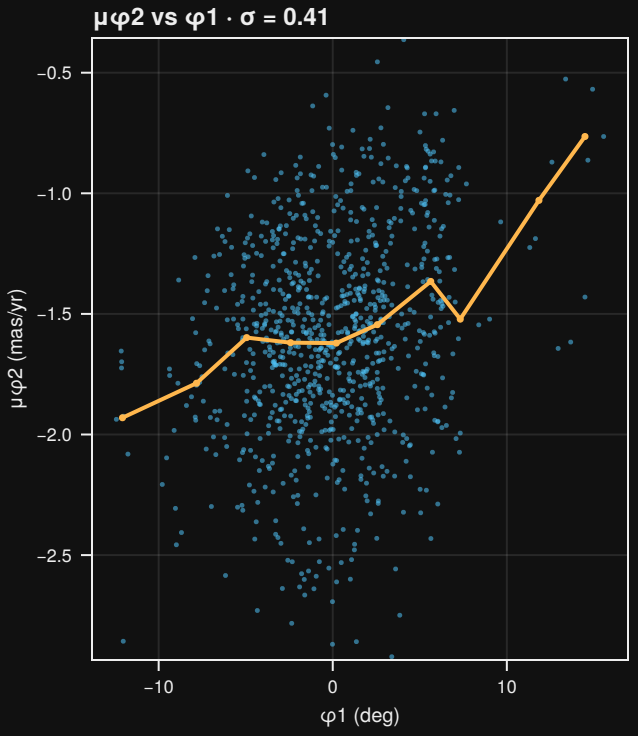


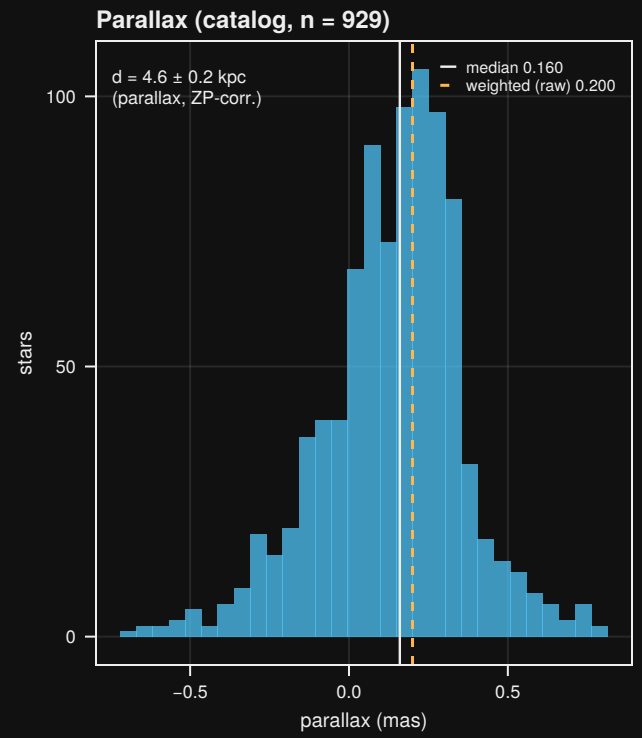
C-12

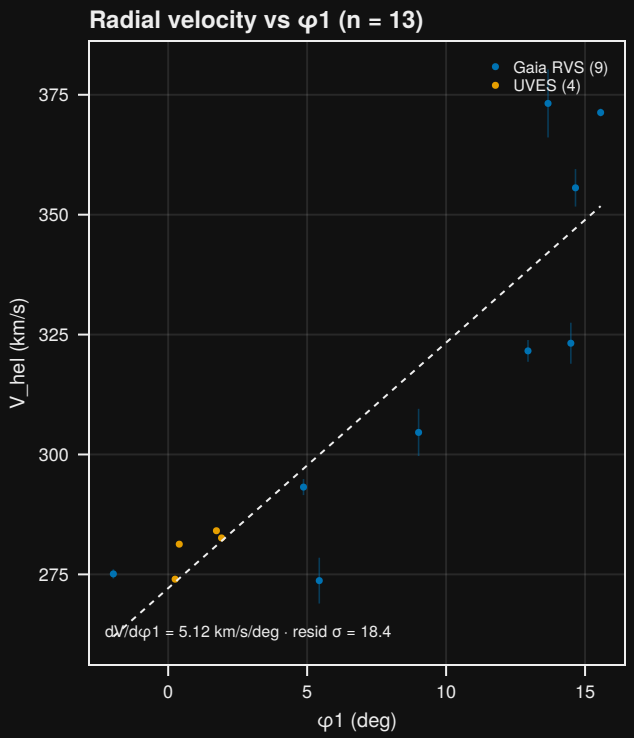
Kinematics · proper motion · parallax · radial velocity

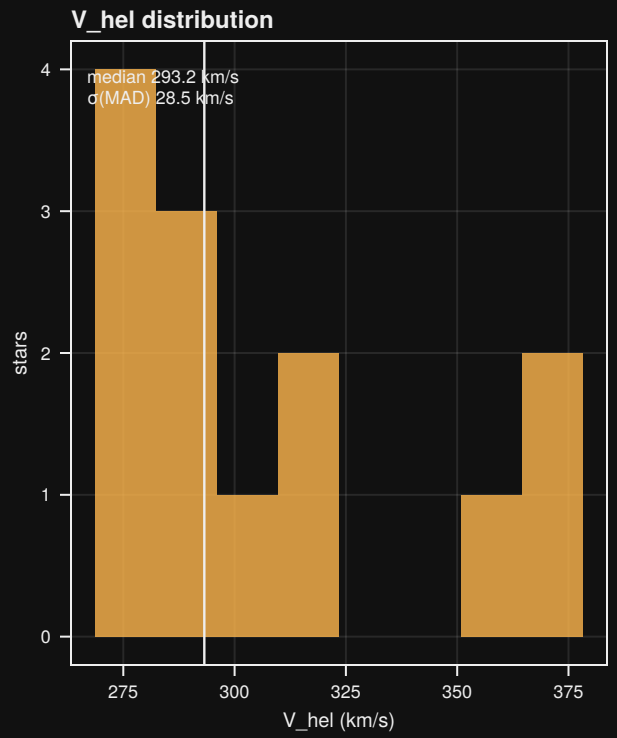






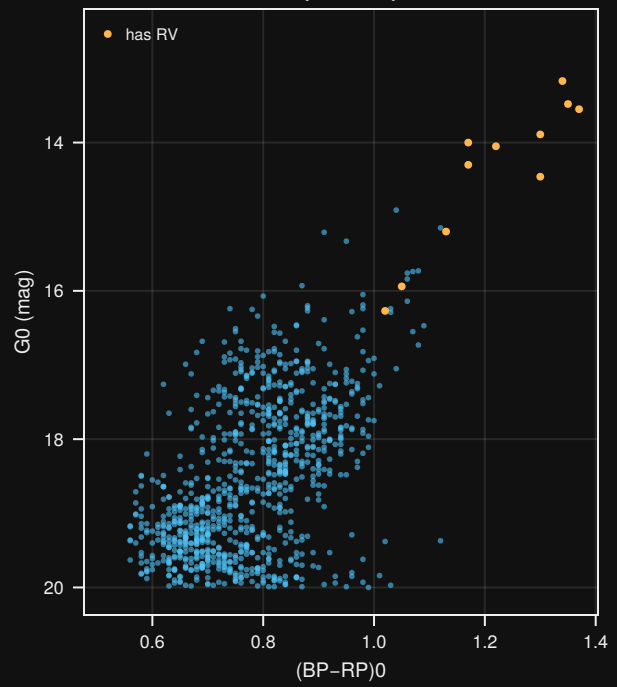




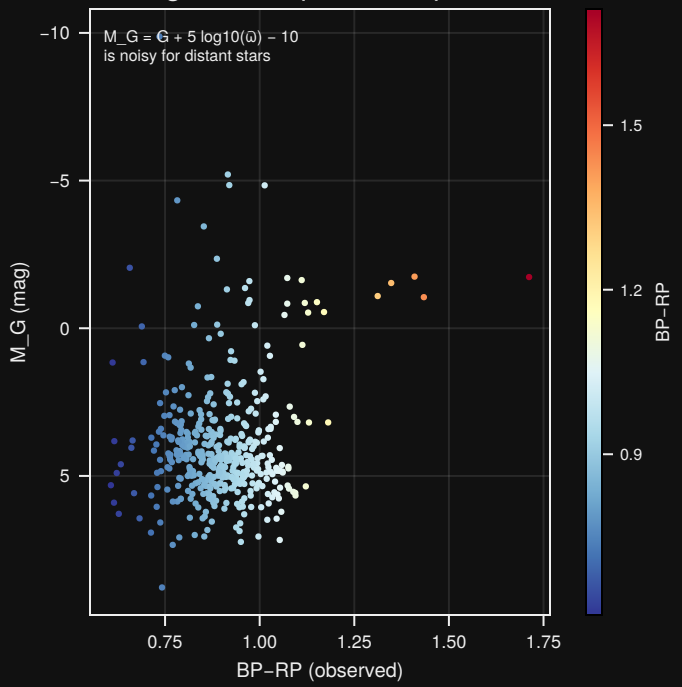


C-12

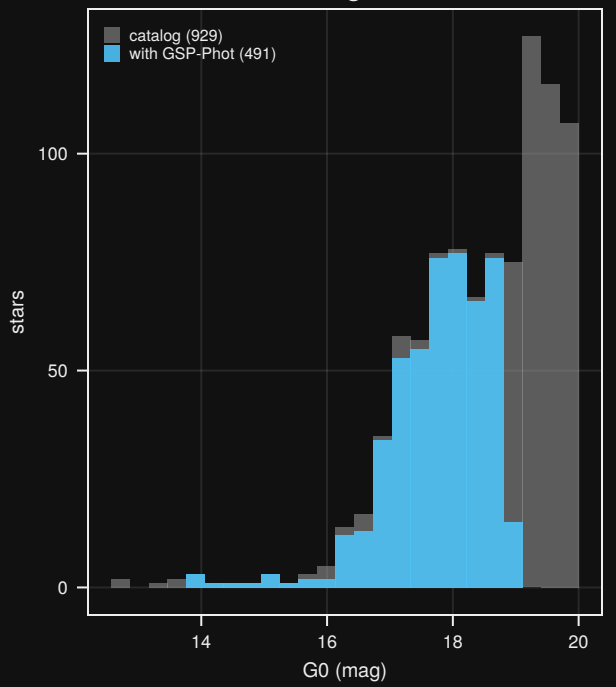
CMD · dereddened (n = 929)



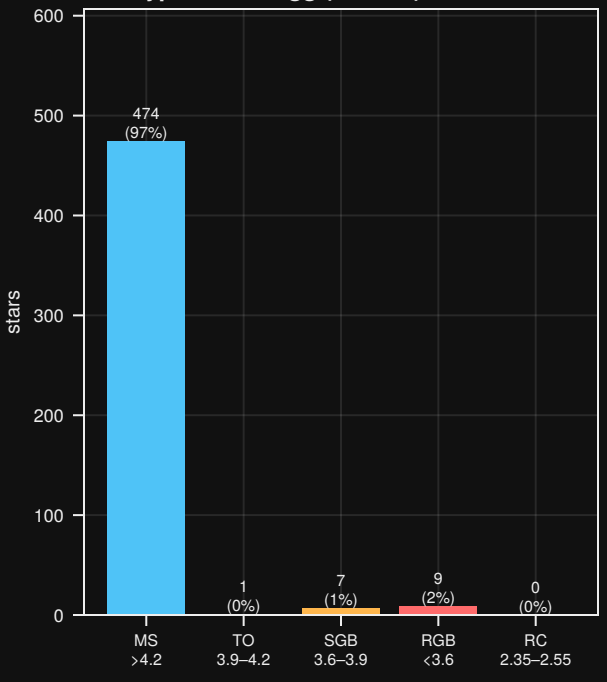
HR diagram · ϖ>0 (n = 445/491)



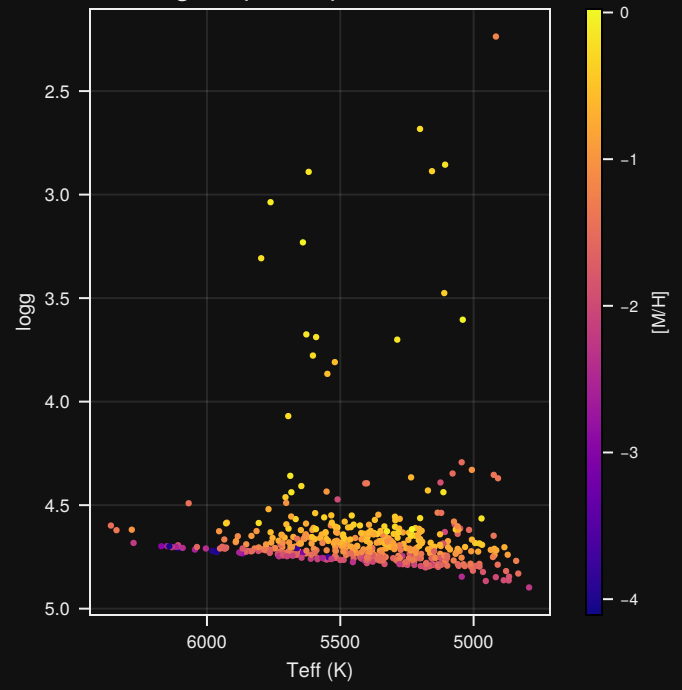
G0 & GSP-Phot coverage



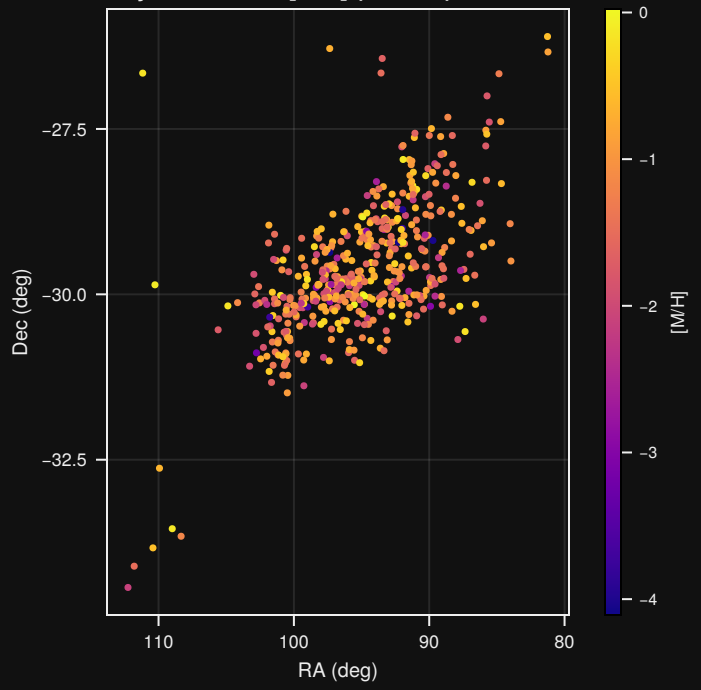
Star types from logg (n = 491)



Kiel diagram (n = 491)



Sky · GSP-Phot [M/H] (n = 491)

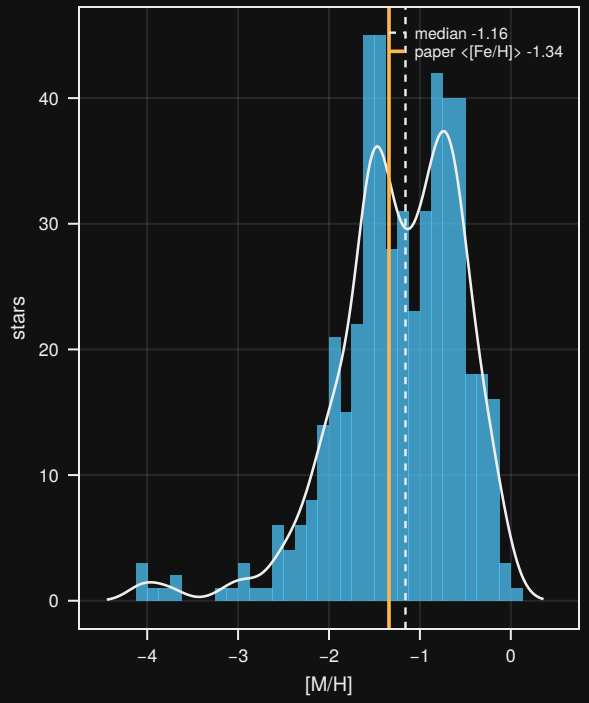


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

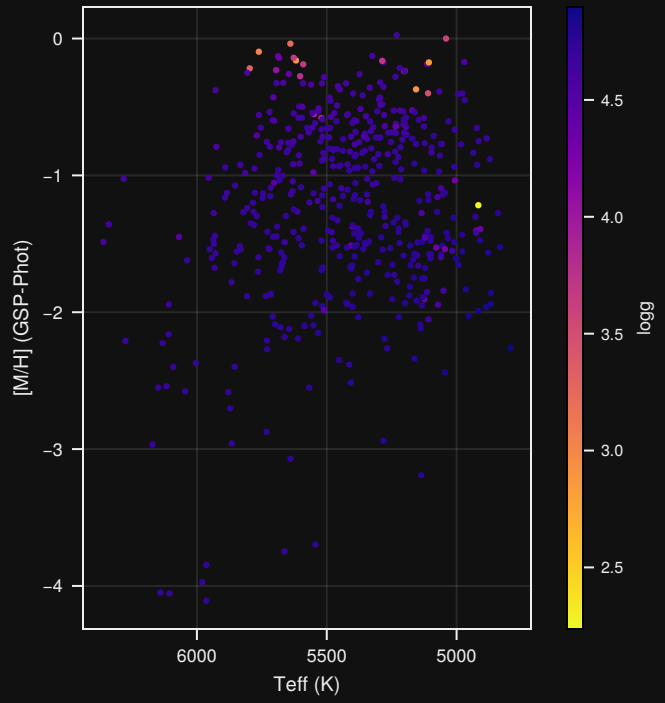
C-12

Metallicity · age · astrometric quality

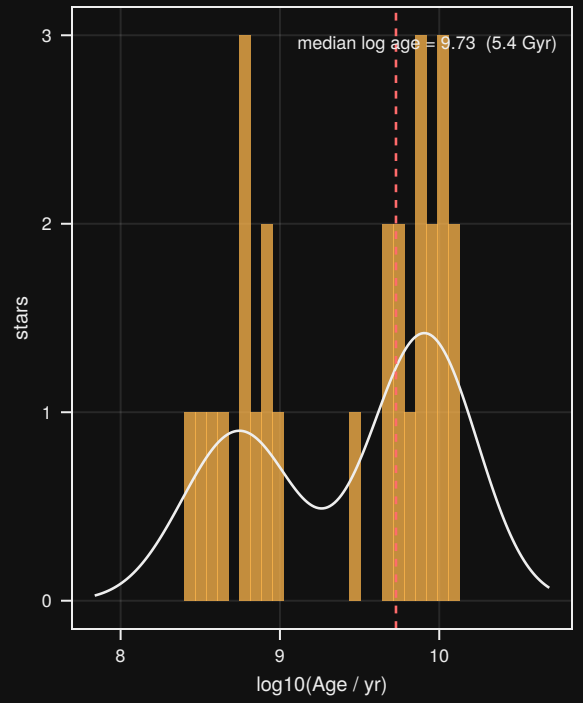
[M/H] GSP-Phot (n = 491)



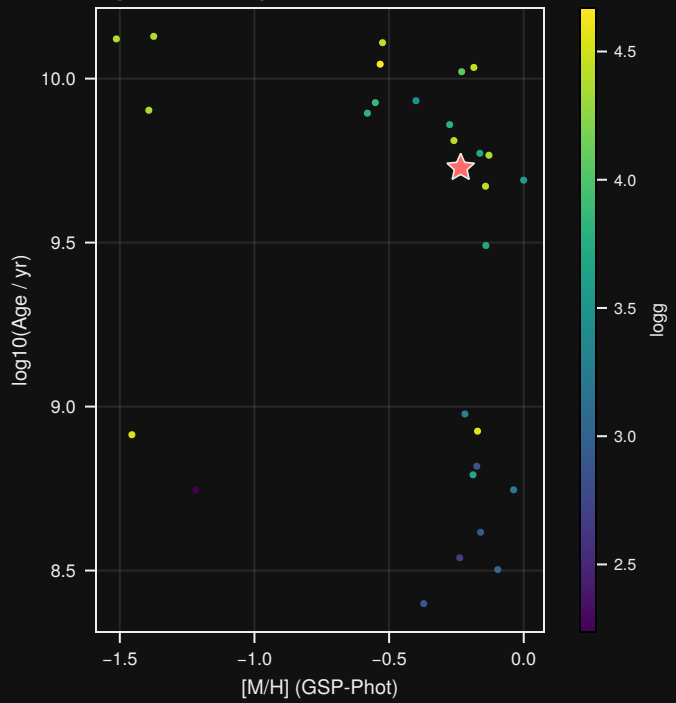
Teff vs [M/H] (n = 491)



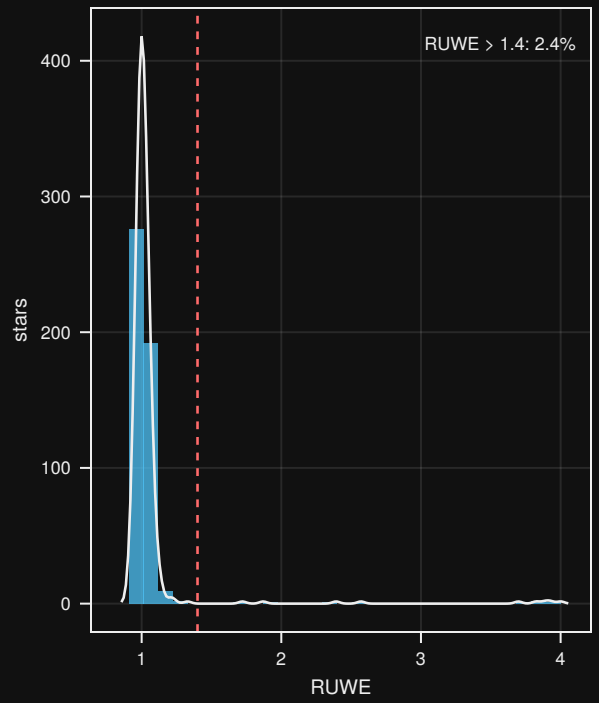
FLAME age (n = 28)



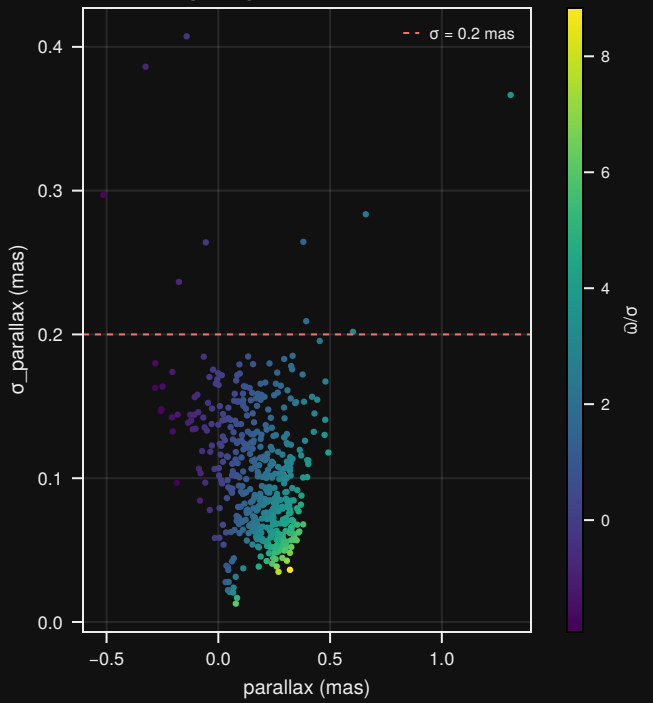
Age–metallicity (n = 28)



RUWE (n = 488)

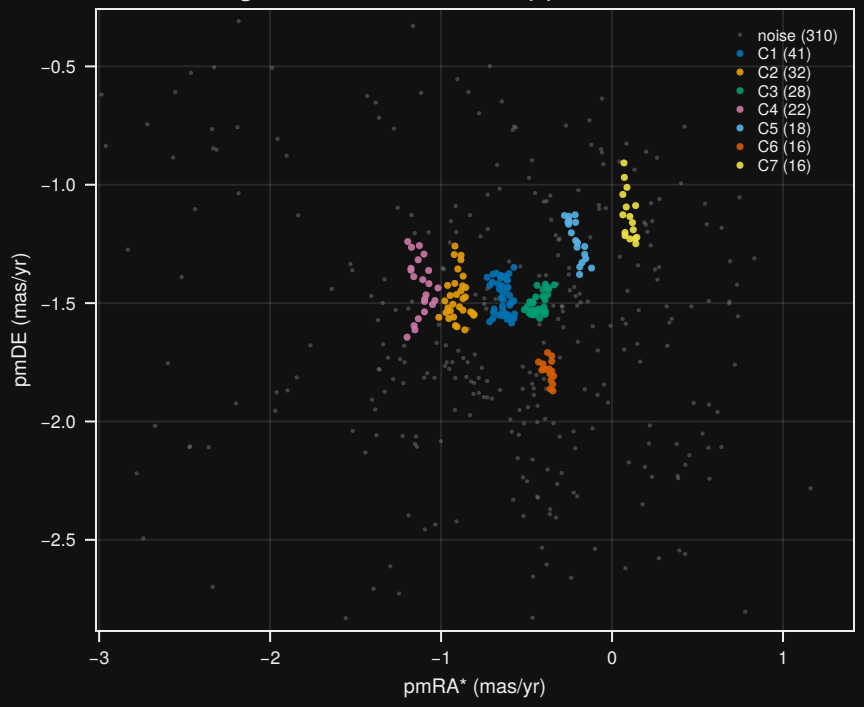


Parallax quality (n = 491)

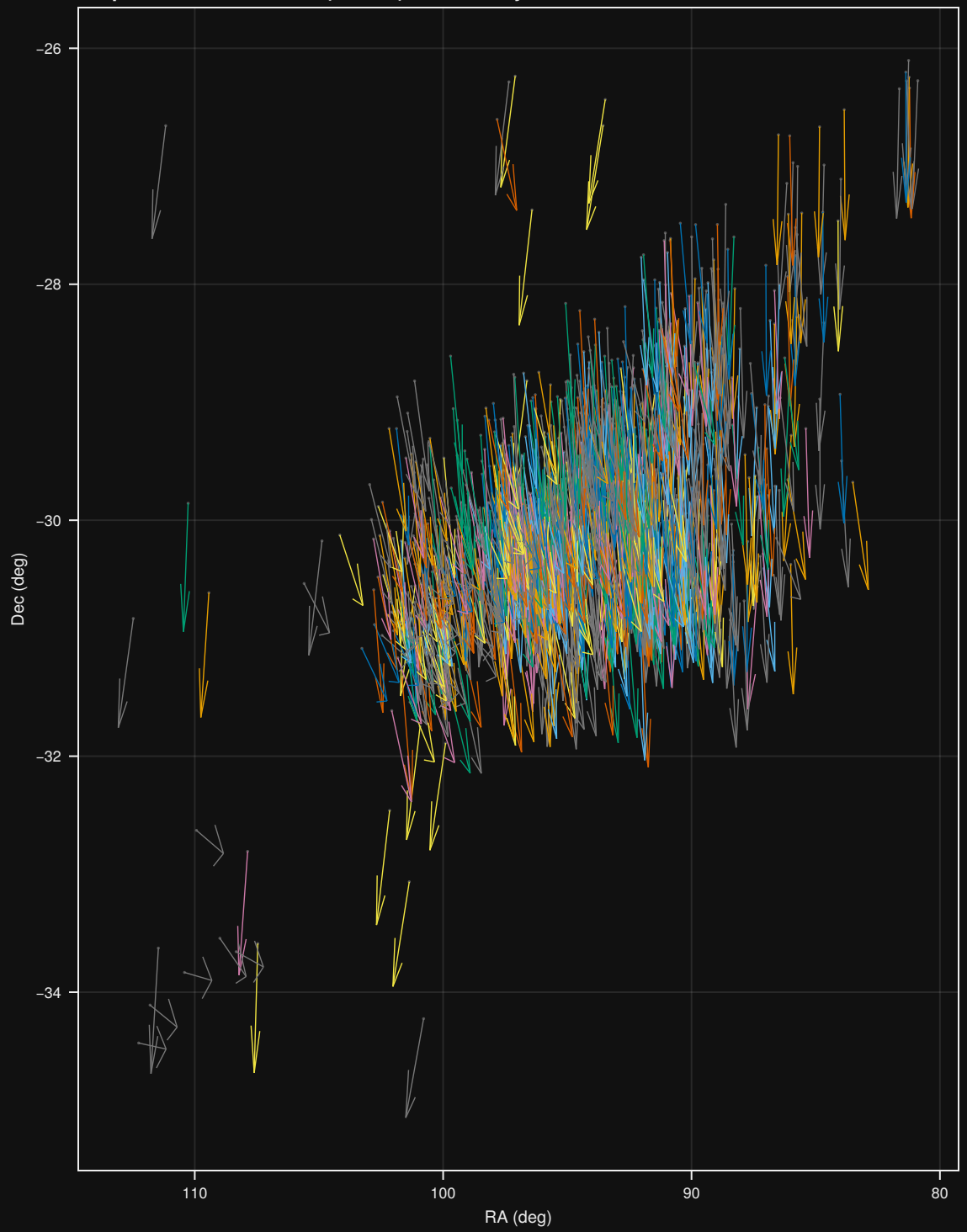


C-12

PM clustering · HDBSCAN · 60 cluster(s) · noise 33%



Proper-motion directions (arrows) coloured by cluster



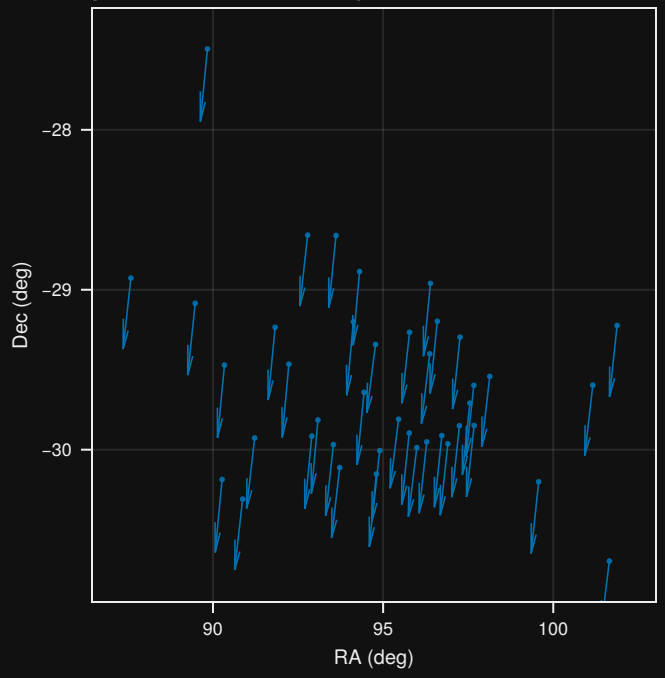
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	310	-0.52	-1.60	0.77	-1.09	5
C1	41	-0.64	-1.48	0.04	-1.47	1
C2	32	-0.90	-1.50	0.05	-1.40	2
C3	28	-0.43	-1.52	0.05	-1.48	1
C4	22	-1.12	-1.43	0.06	-1.17	0
C5	18	-0.21	-1.24	0.06	-0.93	0
C6	16	-0.36	-1.78	0.02	-2.21	0
C7	16	0.10	-1.15	0.04	-1.13	0
C8	16	-0.51	-1.42	0.03	-1.79	0

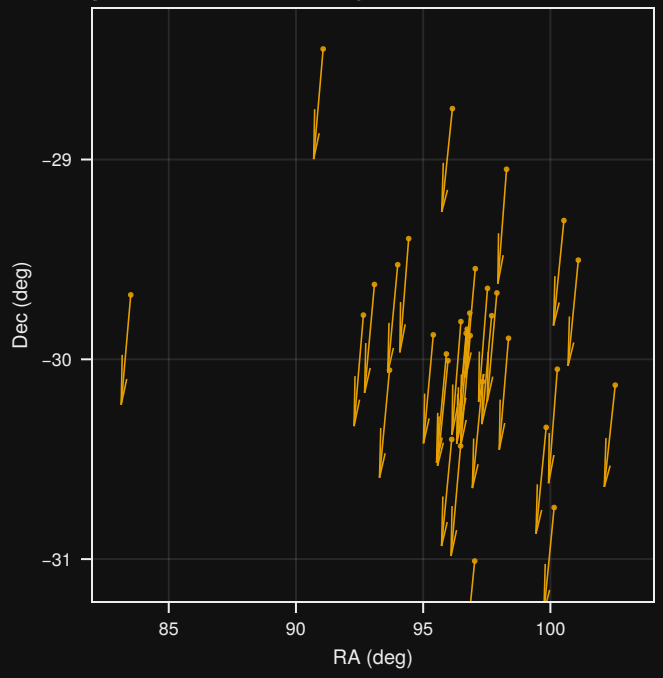
C-12

Per-cluster proper-motion vector maps

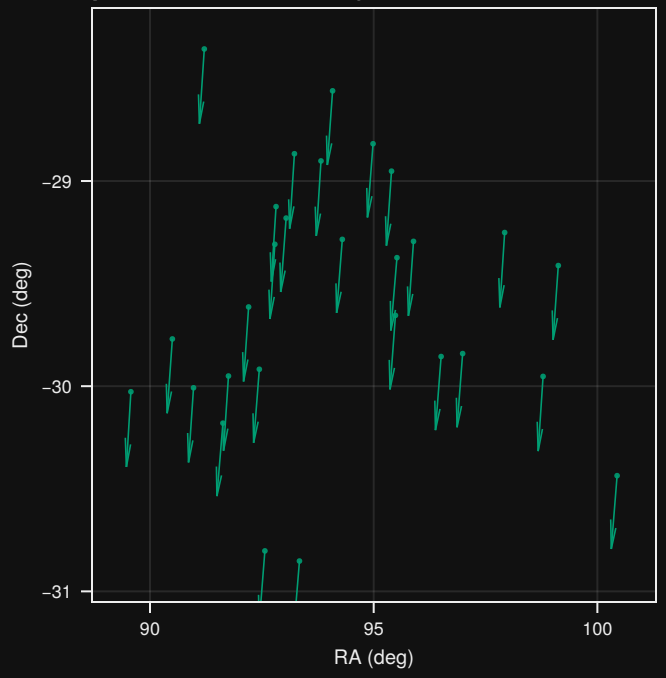
Cluster C1 · n = 41
pm = (-0.64, -1.48) mas/yr



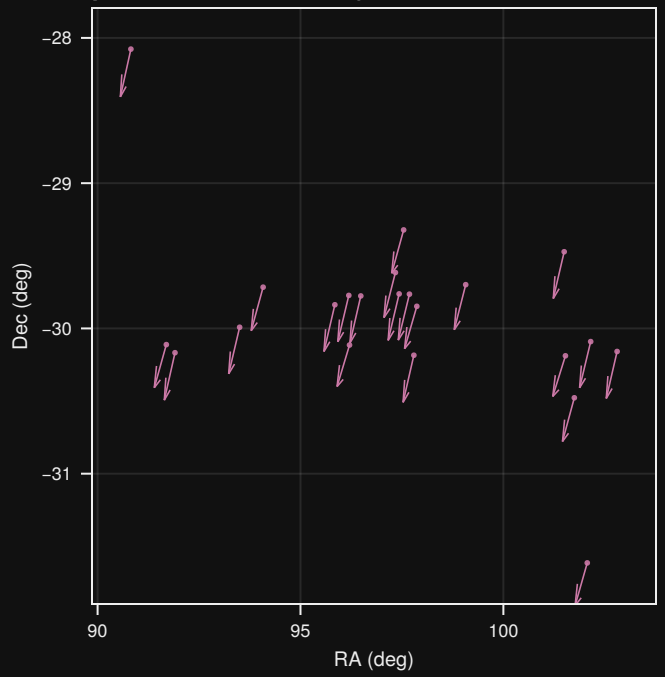
Cluster C2 · n = 32
pm = (-0.90, -1.50) mas/yr



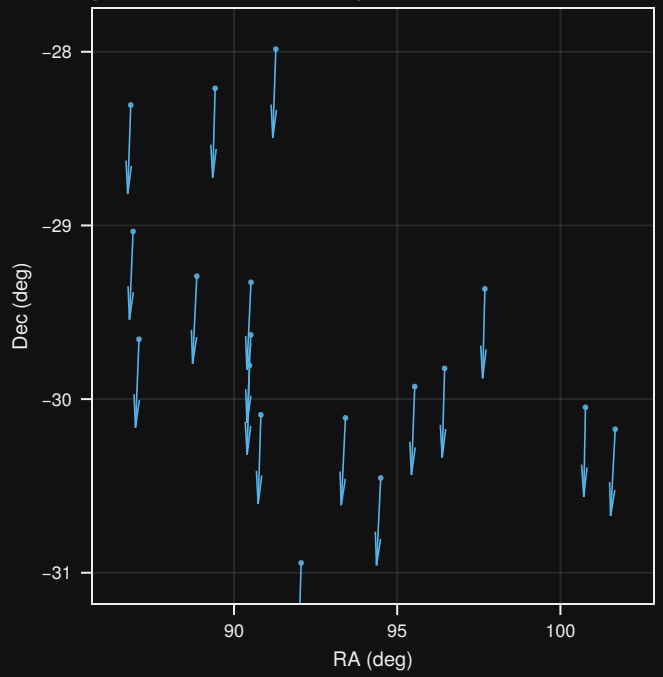
Cluster C3 · n = 28
pm = (-0.43, -1.52) mas/yr



Cluster C4 · n = 22
pm = (-1.12, -1.43) mas/yr



Cluster C5 · n = 18
pm = (-0.21, -1.24) mas/yr



Noise · n = 310

